

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1464

Wayne Trace 100 kV
September 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1464. The Interconnection Customer proposes a 64.6 MW solar photovoltaic facility interconnecting at the Wayne Trace 100 kV substation in the Indiana area. The request was received on April 23, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 482 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,353,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Muldoon Mill / Union-Texas 100.0 kV: 145.6% loading (109 MVA) following the outage of the Montpelier Electric Generating Sta / Muskogee Substation 500.0 kV; DFAX = 7.3%.

Ostrander / Union-Texas 100.0 kV Line: 132.7% loading (107 MVA) following the outage of the Muskogee / Montpelier Electric Generating Sta 500.0 kV; DFAX = 7.3%.

The Ostrander / Peno Crk 100.0 kV line is overloaded to 116.4% of its 115 MVA emergency rating for the loss of the Muskogee / Montpelier Electric Generating Station 500.0 kV L. The project contributes 7.3% DFAX (4.7 MW) to this overload.

The Decatur / Peno Crk 100.0 kV is overloaded to 106.8% of its 118 MVA emergency rating for the loss of the Muskogee / Montpelier Electric Generating Sta 500.0 kV. The project contributes 7.3% DFAX (4.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muldoon Mill / Union-Texas 100.0 kV	Rebuild line with higher-capacity conductor	\$2,623,000	100% (\$2,623,000)
Ostrander / Union-Texas 100.0 kV Line	Rebuild line with higher-capacity conductor	\$3,111,000	100% (\$3,111,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ostrander / Peno Crk 100.0 kV line	Rebuild line with higher-capacity conductor	\$5,757,000	100% (\$5,757,000)
Decatur / Peno Creek 100.0 kV	Reconductor line	\$1,862,000	100% (\$1,862,000)
	Total allocated to project		\$13,353,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.